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M^{ODULE}

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Integration Techniques

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To the Instructor

0.1 Overview of the Module

This module consists of integration techniques: integration by parts (Section 1.2), trigonometric integrals (Section 1.3), trigonometric substitution (Section 1.4), and partial fraction decomposition (Section 1.5). Additionally, Section 1.1 contains a review of basic integrals, and Section 1.6 contains a mixed list of practice integrals.

The module does not cover the use of tables of integrals, integration using technology, numerical integration, definite integrals, or improper integrals. The module guides students as they develop their own understanding of each method, helps them understand where certain formulas come from and how to use them, has them work through a few examples, and leads them to develop some general results. The module takes at least eight 50-minute class periods to complete.

For the module, we assume students know trigonometry (especially right-triangle trigonometry and trigonometric identities), differentiation, transcendental functions, and basic integration. By ‘basic’ integration, we mean that they know the Fundamental Theorem of Calculus, substitution, and integration formulas such as the power rule, e^x , $\sin x$, $\cos x$, $\sec x$, and $\frac{1}{1+x^2}$ (the complete list is on the first page of the module). The students do not need to have seen the more advanced integration techniques before working through the module, nor do they need to have done numerical integration, applications of integrals, or improper integrals.

The sections could be done in almost any order, keeping the following in mind: Section 1.1 is a warm-up and should be done first, Section 1.6 is practice and should be done last, and Section 1.4 should be done sometime after Section 1.3 (there are a couple of exercises in Section 1.4 that require ideas introduced in Section 1.3 - see instructor notes for more details).

0.2 Development and Past Use

This module was developed for our Calculus II course, part of our three-semester calculus sequence. The module was first drafted during the fall of 2015, and since then has been used by at least four different instructors at two different institutions and in conjunction with at least three different calculus textbooks. As of the summer of 2020, the two authors have used the module for a total of 15 sections of Calculus II over the course of seven semesters and one 6-week summer course. We have had a total of 389 students, with sections ranging from 13 to 36 students, and an average of 26 students per section.

As can be expected with four different instructors, the notes have been used in a variety of ways. We will describe the setting at Gonzaga University, where the module was developed and used extensively, as well as how the two authors have implemented it in their classes. The module is presented here without space for students to work on the exercises; however, the instructors print copies of the module for the students that are formatted to include space to complete the exercises. These, as well as the LaTeX source code, are available upon request.

0.2.1 At Gonzaga University

Our Calculus II course consists of 50-minute classes meeting four days per week for a total of approximately 52 days per semester. Excluding review and exam days, this results in roughly 44 days worth of material. The material in Calculus II consists of integration techniques, applications of integration, sequences and series, power series and Taylor series, parametric equations, and polar coordinates.

Calculus II is taken mainly by engineering, math and physics majors in their first year of college. The students generally have taken some calculus in high school, often AP calculus. Some have seen the material in the module previously, and some have not.

Some semesters, the integration techniques module was the first topic covered in the semester. Other semesters, we covered applications of integration first, followed by integration techniques. Both approaches worked well. In the next few sections are details about how the two authors use the module in their courses.

0.2.2 Vesta's Courses

I use modules for every topic in Calculus II, including this one. During class, my students work in groups (more on the groups below) on a portion (or section) of a module. Most days start with a short, five-minute quiz

on the work from the previous day, followed by a discussion of the quiz problem. I then give a very brief introduction to the new section.

The students spend most of the class time working through the sections in their groups. They are expected and encouraged to discuss the questions with each other until the group comes to a consensus. When there are questions, I provide hints, encouragement, prodding in a certain direction, or even a straight answer, depending on what is best for the student's learning. Sometimes, I'll facilitate a whole-class discussion of the question.

I often have students report their answers to the class. This could be done verbally, by presenting on the board, or by filling in responses on a blank copy on the document camera. Another strategy is to have each group send a delegate to report to another group, then return to their own group with any feedback, suggestions, or disagreements. I will occasionally (but only as needed) work through a problem on the board, or give a brief lecture on a particular idea or proof.

Throughout the class period, I am quite mindful of timing, carefully balancing the need to progress through the material with the students' need to think and learn. The goal is to have the students work through the section, or a portion of the section, during class time.

If the pace is too slow, I will assign certain small portions of the module to be done as homework. I then follow up the next day, having students report their solutions to the class. It is important that, by the time we're done discussing a section, students are confident that they understand what they've done and that the solutions they have on their worksheets are correct. But they need to be in class to do the work and come to that confidence - I rarely share with the students my own solutions.

Groups are formed on the first day of class, and students work every day in groups. The groups consist of three or four students. I change the groups every two weeks during roughly the first half of the semester. Sometimes the students choose their own groups, sometimes the groups are arranged so that students are in an entirely new group, and sometimes groups are assigned by myself in some strategic way. For the second half of the semester, I let them choose their own, permanent group.

Initially, the groups tend to be a bit chaotic. So, in order to help the groups function well, each group member is assigned a specific role (changing each day). For example, one student is the presenter, the member who reports to the class that group's solution when called on. Other common roles include a reader to read the instructions out loud, a recorder to ensure the group has a written record of their work, and a consensus builder to ensure the group comes to a consensus before they move on.

Another way I support the functioning of the groups is to occasionally ask the groups to discuss what their group is doing well, or what they are

struggling with and what strategies might help. I then have the groups share their responses with the class. This also gives me a chance to talk about good group etiquette, if needed. Eventually, the groups are functioning well without any assistance from me. I learned about this approach to group work from the POGIL Project (www.pogil.org).

With students working in groups and presenting at the board, the classroom configuration can be critical. However, our classrooms are pretty standard, and I find ways to make them work. Generally, I have one or two large white-boards at the front of the room, together with a document camera and overhead projector. The students sit either at desks or at tables. They quickly learn to rearrange the furniture into groups of four before classes starts, and to return them to the standard rows when class is over.

In addition to the in-class quizzes and modules, I also assign practice problems from a textbook, though they are not collected or graded. We are required to use a department-chosen textbook for our calculus courses, so the students already have the book. The module certainly does not provide enough practice for the average student, so another source of practice problems is a must. Additionally, some semesters, I also assign a small number of practice problems (roughly five per topic) through an online homework system such as WebWork. These problems are graded and contribute to the course grade.

The course grade generally consists of 100 points for quizzes (or 50 points for quizzes and 50 points for homework), 100 points for each of four in-class exams, and 200 points for the two-hour final exam. Even though I have given no points for work on the in-class modules, the students rise to the challenge, work diligently on them during class, and consistently comment on their value on course evaluations. Of course, if student buy-in might be an issue, assigning a participation grade, presentation grade, or module completion grade could be motivational without too much extra work for the instructor.

0.2.3 Katharine's Courses

I also use modules similar to this one for each topic in Calculus II. Most of the class time is spent having students work in groups on a given section of a module. The other details have changed over time to fit the needs of my students. I like to begin each day by asking students if they have any questions remaining from the previous day. Sometimes this leads to a group discussion of something that wasn't understood or that not all students had enough time to work on. Other days, they have no burning questions and we move on quickly.

As students are working on the module, I walk around the room, pausing with each group to check on their progress. I try not to answer questions

they don't ask, but I also try to ensure that I don't see and ignore any wrong answers. When students do have questions, my most frequent responses are pointing at something important on the page (typically directions they've missed or a step they did and forgot about) or asking them a question in response. When several groups have the same question, I quickly pause the entire class so we can have a larger discussion about the issue at hand. I am more prone to pointing students in a good direction than handing them answers, although I will hand out answers and solutions as necessary.

At some points in some modules, I know that I will need to provide a mini-lecture to get the students going. I also start each new section of the modules with a quick (typically just a few sentences) description of what we're doing, how it connects to what came before, and what our bigger picture goals are for the module/course. Periodically, I interrupt the students in order to ask them to share their answers with the class either orally or on the board.

I try not to send students home with portions of a section to complete, but sometimes I do ask them to *attempt* to complete a specific portion (typically one or two items at most) before the next class period. This allows me to leave lingering questions at the end of the day and answer them after the students have had a bit more time to think about the issues. For this module in particular, I tend to leave the last portion (Section 1.6) for the students to work on, either as they're finishing up their work on Section 1.5, or on their own at home.

I am less intentional about the size and changing of groups than Vesta. I encourage students to work in groups of 3 or 4; however, I know that some students prefer to work alone or with just one other person and others have several friends in the class and don't want to leave anyone out. When students working in smaller groups get stuck, my most frequent recommendation is that they ask a nearby group for assistance. Early on they sometimes struggle with this but I find that as the semester continues, those students will do this on their own. The larger groups have a tendency to have subgroups form in twos or threes and these subgroups regularly check in with each other.

I have used daily in-class quizzes; however, I struggled with writing a syllabus that allowed for me to not give a quiz on days when it was less appropriate, and so I have switched that portion of the students' grades to assigned homework. I carefully choose one representative problem per section, give the students the problem on a printed page along with the rubric, and have them turn that in about a week after we discuss the material in class. In the past year, I have also added required practice problems from the textbook to ensure students are getting enough practice with the material.

In addition to homework assignments, I periodically ask my students to

reflect on their learning and the learning process. These assignments were in general inspired by assignments written by Stan Yoshinobu at CalPoly, and include asking students to learn and think about active learning, productive failure, growth mindset, feeling stuck, and their own experiences as learners of mathematics.

The course grade is usually based upon total points accumulated, with 70 points for the reflections, 10 points for each of the 25 homework assignments, 6 exams each worth 80 points, and a final cumulative exam worth 200 points. This gives a total of 1000 points for the semester, which makes grades based on percentages easy to compute. I give many exams in order to allow students to struggle with some topics or have a bad exam day and still do well in the course overall. This also lessens the amount of new content per exam, which means that I'm less likely to write exams that are too long and too hard (making them each out of 80 points also helps with this - I'm all in favor of any and all tricks that work for you!). In addition to the graded components of the course, I also give my students a list of problems from the textbook that I recommend they do for additional practice.

Asynchronous Online

The summer 2020 iteration of this course was mostly a successful experiment. Due to COVID-19, the 6 week course was delivered asynchronously and online. The students were given this module (and the others we have developed) to work through on their own or with classmates. The only video created for this module was the suggested example from right before Exercise 28. In this video, I purposefully did a bit of a guess-and-check strategy, and made some poor choices, so that I could explain how I chose the recommended substitution choice. There were traditional written and graded homework assignments, three non-cumulative exams, and a variety of ways for students to earn participation points by engaging with the course material.

I stressed working with their classmates by making that activity worth more participation points than any other participation activity. Completion of each module section was also worth a substantial number of points. The strategy of this choice was to encourage them to learn from the modules rather than directly from the textbook, or relying on internet sources that had not been vetted by me. Students could also earn participation points by asking and answering questions on the online platform Piazza, finding errors in my typed materials, asking me questions in virtual Zoom office hours, and the occasional non-academic task (working outside over the Memorial Day weekend was very popular). I say this experiment was successful because almost all of the students worked with their peers and learned sufficiently well to pass the course. Additionally, most of the students completed the

modules for participation points and found that to be the most effective way to learn in our necessarily online and asynchronous class.

0.3 Nuts and Bolts

The introduction above is heavy on overall course structure and light on details about how to actually use this module in the classroom. In this Instructor Version of the module, you will find many additional comments throughout the body of the module. These include descriptions of how we scaffold, introduce, and implement the module during our in-class time, hints we recommend giving to students who are stuck, and answers for all integrals.

To the Student

The purpose of this module is to guide you as you learn integration techniques, to understand how they work and how to use them. You will be reading explanations as well as working through exercises.

As you read, if you are not completely understanding the mathematics, stop and read it again. It might help to read out loud yourself, or have someone else read out loud while you listen. Write down the calculations as you read. Make sure you understand each step and each statement before you go on. If you are not sure about a particular point, discuss it with your classmates or ask your instructor.

As you work exercises, it is likely that you will struggle or become stuck at some point. Consider the poem “Our Real Work” by Wendell Berry:

It may be that when we no longer know what to do
we have come to our real work,
and that when we no longer know which way to go
we have come to our real journey.
The mind that is not baffled is not employed.
The impeded stream is the one that sings.

Allow yourself to struggle. When you are stuck, try something else. Reread or rework the material leading up to where you got stuck. Make sure you do not have a previous mistake that is leading to your stuck-ness. Try something different. Discuss with your classmates an alternative approach. Whatever you do, recognize that your real work is in these moments when you are struggling or stuck. Practice persistence. Ask questions. Keep trying until you get it figured out - the process is as important as reaching a correct solution.

When you finish with a section of this module, you are not done. You then need to practice the techniques you have learned by working through a bunch of homework problems.

So, read carefully, persist when you are struggling, ask questions, practice, and enjoy!

Chapter 1

Integration Techniques

1.1 Review of the Basics

Begin by motivating the need for integration techniques. If students have seen some applications of integration, where they probably encountered functions they could not integrate, remind them of this. Otherwise, talk about some possible applications (they probably know that definite integrals can measure area under a curve, but you can also hint at other applications such as volume and work) and explain that these will produce functions that require more advanced integration techniques. However, we begin with a set of warm-up exercises to review the formulas and techniques (specifically, u -substitution) that they already know.

The integration formulas the students should know are listed, excluding some very basics ones like constant multiple rule. All the integration formulas needed for examples and exercises throughout the module are included in the list. There are lots of review exercises using the formulas and substitution. Some require algebraic techniques, which are given in the hints. Some require trigonometric formulas.

These exercises could be worked in class, in part or in their entirety. They could be assigned as homework and checked in class. Etcetera. The exercises help the students polish their algebra and integration skills, and prepares them for the more advanced integration techniques.

(Basic Integration Rules.)

$$\int c \, dx = cx + C$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

$$\int x^{-1} \, dx = \ln|x| + C$$

$$\int e^x \, dx = e^x + C$$

$$\int \frac{dx}{1+x^2} = \arctan x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \csc x \cot x \, dx = -\csc x + C$$

You should know the integration rules in Key Idea 1.1. If there are any of the rules that you are not comfortable with, please ask! If there are any basic rules that are missing from the list, add them in. Additionally, you are expected to know basic substitution. Exercise 1.1 will help you review these integration rules, as well as some useful algebraic techniques.

Calculate each integral.

Some exercises, such as this one, ask the students to DO quite a bit of work. In other exercises, the main purpose is that the students are READING and UNDERSTANDING.

Some of the exercises will be challenging for students because they require either substitution or algebra (for example, long division). Have a plan for addressing their struggles. One option is to review substitution and maybe some algebraic techniques such as long or synthetic division and completing the square first, then let the students work on the exercises. Another option is to let them work and struggle first, giving them opportunities to share their solutions with the class, then review substitution and algebra as needed. Answers, and some additional hints, are provided after each exercise. Specifically for polynomial division, it helps to first do an example of (regular) integer long division to demonstrate the similarities - talk about place value and make the analogy that $7x$'s is like 7 tens.

a. $\int \frac{1}{2-3x} \, dx$

Answer: $-\frac{1}{3} \ln|2-3x| + C$

b. $\int \sec^2 x \, dx$

Answer: $\tan x + C$

c. $\int \tan \theta d\theta$

Hint: $\tan \theta = \frac{\sin \theta}{\cos \theta}$

Answer: $-\ln |\cos \theta| + C = \ln |\sec \theta| + C$

d. $\int e^{\sin x} \cos x dx$

Answer: $e^{\sin x} + C$

e. $\int (1 + \tan^2 x) dx$

Hint: $1 + \tan^2 x = \sec^2 x$

Answer: $\tan x + C$

f. $\int \frac{x^2 - 2x - 3}{x + 1} dx$

Hint: factor and reduce

Answer: $\frac{1}{2}x^2 - 3x + C$

g. $\int \frac{\ln x}{x} dx$

Answer: $\frac{1}{2}(\ln x)^2 + C$

h. $\int \sec x \tan x dx$

Answer: $\sec x + C$

i. $\int \frac{x}{\sqrt{1 + 4x^2}} dx$

Answer: $\frac{1}{4}\sqrt{1 + 4x^2} + C$

j. $\int \frac{e^t - e^{-t}}{e^t + e^{-t}} dt$

Hint: $u = e^t + e^{-t}$

Answer: $\ln(e^t + e^{-t}) + C$

k. $\int \frac{1}{e^x + e^{-x}} dx$

Hint: first multiply by $1 = \frac{e^x}{e^x}$, then use a substitution.

Hint: $u = e^x$

Answer: $\arctan e^x + C$

l. $\int \frac{1}{\sqrt{9 - x^2}} dx$

Hint: first use $\sqrt{9 - x^2} = 3\sqrt{1 - (x/3)^2}$, then use a substitution.

Hint: $u = \frac{x}{3}$

Answer: $\arcsin \frac{x}{3} + C$

m. $\int \frac{3x+1}{\sqrt{1-x^2}} dx$

Hint: first split up the fraction using the rule that $\frac{a+b}{c} = \frac{a}{c} + \frac{b}{c}$.

Answer: $-3\sqrt{1-x^2} + \arcsin x + C$

n. $\int \frac{x+1}{-x^2+3x+4} dx$

Hint: first reduce the rational function (factor the denominator).

Answer: $-\ln|x-4| + C$

o. $\int \frac{x^2+2x-1}{x+4} dx$

Hint: first use long division to simplify the improper rational function.

Hint: students may need to be reminded of how to rewrite the rational function after doing long or synthetic division;

Answer: $\frac{1}{2}x^2 - 2x + 7\ln|x+4| + C$

p. $\int \frac{1}{x^2+8x+17} dx$

Hint: first complete the square, then use a substitution.

Answer: $\arctan(x+4) + C$

1.2 Integration by Parts

Perhaps begin with a mini lecture setting the stage for developing and using integration by parts. Point out that, for example, we cannot calculate $\int x \sin x dx$ using tools developed so far. Notice, though, that the integrand is a product. While we do not have a product rule for integration - we cannot write

$$\int x \sin x dx = \left(\int x dx \right) \left(\int \sin x dx \right)$$

(you could have the students check that the right side does not give the correct answer by integrating then differentiating) - maybe we can get some ideas from the product rule for differentiation: $\frac{d}{dx} fg = fg' + g'f$. Explain how to write the product rule using differentials, $d(uv) = u dv + v du$, introducing them to the notation used in Exercise 1.2. Next, have them read through the development of the integration by parts formula in Exercise 1.2. One reason for having the students read and work through the development, rather than just including it in the mini lecture, is to further develop their ability to read mathematics - if they are doing this in class and together, it's less daunting and they really improve. They may also learn to think more

mathematically, for themselves. Alternatively, you may choose to talk them through Exercise 1.2 as part of your mini lecture.

Also note that on some integrals, depending on choices made during integration, the answers can differ by a constant and still be correct. This is not always immediately obvious.

First, let's see how we can come up with an integration formula using the product rule for differentiation. As you read through, make sure you understand each step. Start with the product rule for differentiation of uv , and write it in differential notation:

$$d(uv) = u dv + v du.$$

Integrating both sides of the product rule gives

$$\int d(uv) = \int (u dv + v du) \quad (1.1)$$

$$uv = \int u dv + \int v du \quad (1.2)$$

Question: What rule or theorem allows us to simplify from equation 1.1 to equation 1.2?

The answer is essentially the Fundamental Theorem of Calculus and the sum rule for integration.

Solving equation 1.2 for $\int u dv$ gives

$$\int u dv = uv - \int v du$$

which is the rule for integration by parts.

(Integration by Parts.)

$$\int u dv = uv - \int v du$$

This exercise walks the students through a step-by-step process for using integration by parts. Students really struggle with the freedom to make choices in mathematics. So one goal here is to get students to feel more comfortable making choices, analyzing them, then deciding which might be best. Point out that it's okay if a choice turns out bad, but we should consider why it is bad (learn from it) then make another choice. Encourage students to work through the process of making and trying choices to build intuition, rather than relying on "rules" (for this topic in particular, students may have heard that the order for choosing the part to differentiate uses the acronym LIATE - logs, inverse trig, algebraic, trig, exponential). While rules can be helpful if they're understood well, simply memorizing them for

an exam won't help students outside of a classroom scenario, so we try to deemphasize them.

Some groups of students struggle to read carefully and want to fill in everything at once, ahead of the directions. For these students, we encourage them to not do so by reminding them to only work on the assigned task. Other groups of students seem to not be able to continue until they've completed an assigned task and feel confident that they've done it correctly. For both groups, we find that the occasional answer in the notes is helpful, either in checking their work, or building confidence that they're making good progress.

Integrating by parts requires a number of steps, and sometimes trial and error. We will go through the steps to integrate $\int x \sin x dx$.

Step 1: When using the integration by parts formula, we first need to choose u and dv so that our integral looks like $\int u dv$. For the integral in the table below, make two different choices for u and dv . *Note:* Initially, don't worry about whether the choices are good or not as long as $u dv = x \sin x dx$. For now, you need only choose u and dv ; we will fill in the other columns in the tables in later steps. I've made the first choice for you.

$\int x \sin x dx$	u, dv	du, v	Integration by Parts
First Choice	$u = \sin x$ $dv = x dx$	$du = \cos x dx$ $v = \frac{1}{2}x^2$	$\int x \sin x dx =$ $\frac{1}{2}x^2 \sin x - \int \frac{1}{2}x^2 \cos x dx$
Second Choice			

Step 2: For each choice in the table, calculate du and v by differentiating u and anti-differentiating/integrating dv . Fill in the appropriate column with your results. Again, I've done the first ones for you.

Step 3: Rewrite the integral using the Integration by Parts formula and fill in the final column, like I did for the First Choice.

Step 4: Now we need to analyze the new integral. Mark a choice as GOOD if the new integral is something we can integrate or is nicer than the one we started with. Mark a choice MAYBE if it is at least no worse than the one we started with. Otherwise mark the choice as BAD.

We usually stop here and ask students what their second choice was and why they thought each choice was good, bad or maybe. The idea

is that a choice is good if it makes the new integral appear simpler, or at least no worse, than the original integral. The first choice is a bad choice since the original integral has a factor of x , while the new integral has a factor of x^2 . On the other hand, the other likely choice is

$$\begin{aligned}u &= x \\ dv &= \sin x dx,\end{aligned}$$

which gives

$$\int x \sin x dx = -x \cos x - \int (-\cos x) dx = -x \cos x + \sin x + C.$$

Check that the students are dealing carefully and correctly with the negative signs, which is something they struggle with.

Step 5: Using the GOOD choice, finish calculating $\int x \sin x dx$.

The remaining exercises lead the students through the ‘tricks’ to using integration by parts.

We will use the same process to evaluate $\int x^2 e^x dx$. Start by filling in the table and analyzing each choice as GOOD, MAYBE, or BAD.

$\int x^2 e^x dx$	u, dv	du, v	Integration by Parts
First Choice			
Second Choice			

Now, use the GOOD choice to finish calculating $\int x^2 e^x dx$. You will have to use integration by parts a second time.

The good choice is $u = x^2$. One issue is that they need to make the same pattern of choices for the second round of integration by parts, i.e., the second choice is $u = x$ or some constant times x , depending on their choices for dealing with the -2 coefficient. It is a great learning moment if they switch the pattern of choices, and investigate what happens. Done correctly, they should get something equivalent to

$$\begin{aligned}\int x^2 e^x dx &= x^2 e^x - \int 2x e^x dx \\ &= x^2 e^x - [2x e^x - \int 2e^x dx] \\ &= x^2 e^x - 2x e^x + 2e^x + C\end{aligned}$$

(although parenthesis and constants may vary, depending on algebraic choices along the way). Again, make sure they are dealing correctly with the negative signs, coefficients and parentheses.

For $\int \ln x \, dx$, we cannot let $dv = \ln x \, dx$ since we don't know the antiderivative of $\ln x$. So we must let $u = \ln x$. Determine dv , du and v and apply integration by parts to calculate $\int \ln x \, dx$.

Students are uncomfortable with letting $dv = 1 \, dx$ and finding $v = x$. It is, admittedly, a bit scary. They may need some encouragement. They may also be unsure how to integrate $\int 1 \, dx$. They should get

$$\int \ln x \, dx = x \ln x - \int 1 \, dx = x \ln x - x + C.$$

Now calculate $\int e^x \sin(3x) \, dx$. Start by using the same GOOD or MAYBE choices and integration by parts multiple times.

The algebra 'trick' in this problem is a challenge for students. We usually let them try for a bit to see what's happening if they keep using integration by parts, and what they might do about the problem. They may not figure out the trick for themselves, but you can certainly push them in a good direction using our reminder to you below. You may also choose to give a mini-lecture to explain what to do:

After doing integration by parts twice, and depending on your choices, you should end up with

$$\int e^x \sin(3x) \, dx = \text{expression} + \text{constant} \cdot \int e^x \sin(3x) \, dx.$$

The trick is to let $I = \int e^x \sin(3x) \, dx$ (on both sides of the equation) and solve for I by subtracting $(\text{constant} \cdot I)$ from both sides of the equation. Then divide by the coefficient on the left to solve for the integral $I = \int e^x \sin(3x) \, dx$.

Depending on whether they initially choose $u = \sin(3x)$ (the option using the chain rule to find du , rather than substitution for finding v) or $u = e^x$, they should get something like

$$\begin{aligned} \int e^x \sin 3x \, dx &= e^x \sin 3x - \int 3e^x \cos 3x \, dx \\ &= e^x \sin 3x - [3e^x \cos 3x - \int -9e^x \sin 3x \, dx] \\ &= e^x \sin 3x - 3e^x \cos 3x - 9 \int e^x \sin 3x \, dx. \end{aligned}$$

So $10 \int e^x \sin 3x \, dx = e^x \sin 3x - 3e^x \cos 3x$, and hence

$$\int e^x \sin 3x \, dx = \frac{1}{10} e^x \sin 3x - \frac{3}{10} e^x \cos 3x + C.$$

Be sure to work out for yourself what you get with the initial choice $u = e^x$ so that you can help students who make this choice.

1.3 Trigonometric Integrals

All the needed formulas (Half-Angle Identities, etc.) are given throughout the section, with some of the exercises encouraging the students to take some ownership of the formulas by filling in missing pieces, calculating du for a substitution, etc.

The methods for trig integrals are given: using a Half-Angle Identity followed by a substitution, using a Pythagorean Identity for sine or cosine followed by a substitution, using a Pythagorean identity for tangent or secant followed by a substitution, and encouragement to be persistent for the tangent/secant integrals that don't fit the patterns. At the end of each method, there is an exercise in which the students need to think about the circumstances under which that method will work.

You might begin with a mini lecture. Explain that we now want to integrate powers of trig functions. For example, we can integrate

$$\int \sin x \cos^2 x \, dx$$

using a substitution $u = \cos x$. Have them work it out as a warm-up, or do it yourself if you're short on time. Mention that other integrals such as $\int \cos^2 x \, dx$ and $\int \sin^3 x \cos^2 x \, dx$ require a bit more work and the use of trig identities. Then let them work on the Exercises.

(Half-Angle and Double-Angle Identities.)

Half-Angle Identities	Double-Angle Identities
$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$	$\sin 2\theta = 2 \sin \theta \cos \theta$
$\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$	$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$
	$= 2\cos^2 \theta - 1$
	$= 1 - 2\sin^2 \theta$

Use Half-Angle Identities (see Key Idea 1.3) to calculate $\int \sin^2 x \, dx$ and $\int \cos^2 x \, dx$.

The answers are $\frac{1}{2}x + \frac{1}{4}\sin(2x) + C$ and $\frac{1}{2}x - \frac{1}{4}\cos(2x) + C$ respectively.

Your answers to Exercise 1.3 probably involve the angle $2x$. Use Double-Angle Identities to rewrite your answers to involve only the angle x . This may be useful later when we are doing trig substitutions in Section 1.4.

Changing the angle from $2x$ to x may be used for Exercise 1.4 e. in Section 1.4, depending on choices made while integrating.

The answers are $\frac{1}{2}x + \frac{1}{2}\sin x \cos x + C$ and $\frac{1}{2}x - \frac{1}{4}\cos^2 x - \frac{1}{4}\sin^2 x + C$ respectively.

Use Half-Angle Identities to calculate $\int \cos^2 x \sin^2 x dx$.

Students struggle with the algebra and with getting an angle of $4x$ after the last application of a half-angle identity (they leave the angle as $2x$). The calculation is:

$$\begin{aligned}
 \int \cos^2 x \sin^2 x dx &= \int \frac{1}{2}(1 + \cos 2x) \frac{1}{2}(1 - \cos 2x) dx \\
 &= \int \frac{1}{4}(1 - \cos^2 2x) dx \\
 &= \int \frac{1}{4}\left(1 - \frac{1}{2}(1 + \cos 4x)\right) dx \\
 &\vdots \\
 &= \frac{1}{8}x - \frac{1}{32}\sin 4x + C.
 \end{aligned}$$

Take the time to make sure they all understand and correct any mistakes, either by checking each group, or having a student share their solution with the class (and making corrections if needed), or going through it yourself after they've finished. Or have students or groups compare their solutions with each other to help them discover where they are making mistakes.

(Substitutions and the Pythagorean Identities.)

Substitution	Pythagorean Identity
If $u = \sin \theta$, then $du = \boxed{}$.	$\cos^2 \theta + \sin^2 \theta = 1$
If $u = \cos \theta$, then $du = \boxed{}$.	$\cos^2 \theta = \boxed{}$ $\sin^2 \theta = \boxed{}$

Fill in the the du parts of the two substitutions in Key Idea 1.3. Fill in the missing parts of the alternate versions of the Pythagorean Identity in Key Idea 1.3 by rewriting $\cos^2 \theta + \sin^2 \theta = 1$ in appropriate ways.

Have a student or group report their answers to quickly check that they are starting with the correct formulas.

Calculate $\int \sin^2 x \cos x dx$ using an appropriate substitution from Key Idea 1.3. The answer is $\frac{1}{3}\sin^3 x + C$.

Now calculate $\int \sin^3 x \cos^2 x dx$ in a similar way after first using the Pythagorean Identity to rewrite $\sin^2 x$ in terms of $\cos^2 x$.

You might have to point out that $\sin^3 x = \sin x(\sin^2 x)$, or have students or groups check in with each other for this suggestion. With $u = \cos x$, they

should get

$$\int \sin^3 x \cos^2 x \, dx = \int \sin x (1 - \cos^2 x) \cos^2 x \, dx = \frac{1}{5} \cos^5 x - \frac{1}{3} \cos^3 x + C.$$

Look back over all of the exercises with integrals of the form $\int \sin^m x \cos^n x \, dx$. The approach you take to integrating depends on whether m and n are odd or even. Fill in the missing words in the rules below:

- For $\int \sin^m x \cos^n x \, dx$, if m and n are _____ then use Half-Angle Identities.
- For $\int \sin^m x \cos^n x \, dx$, if m or n is _____ then use the Pythagorean Identity and Substitution with $u = \cos x$ if _____ and $u = \sin x$ if _____.

Have a student or group report their answers. The answers are: even; odd; m odd; n odd. Note that either either choice works if both exponents are odd.

(Tan & Sec Identities and Integrals.)

Identity	Integral
$1 + \tan^2 x = \sec^2 x$	$\int \tan x \, dx = \ln \sec x + C$ $\int \sec x \, dx = \ln \sec x + \tan x + C$

The only new information in Key Idea 3 may be $\int \sec x \, dx = \ln |\sec x + \tan x| + C$. The calculation begins as follows:

$$\begin{aligned} \int \sec x \, dx &= \int \sec x \left(\frac{\sec x + \tan x}{\sec x + \tan x} \right) dx \\ &= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx \end{aligned}$$

Finish the calculation using a u -substitution with $u = \sec x + \tan x$.

We want to use some combination of Key Idea 1.3, substitution, and integration by parts to calculate integrals of the form $\int \tan^m x \sec^n x \, dx$. Whenever possible, use the identity $1 + \tan^2 x = \sec^2 x$ and do a substitution with $u = \sec x$ or $u = \tan x$. For the substitution, you'll need to calculate du . If $u = \sec x$ then $du =$ _____. If $u = \tan x$ then $du =$ _____. When using one of these substitutions, you must 'save' out the du part and then rewrite the rest of the integral in terms of u using the identity.

Calculate each integral.

It can be challenging to figure out what to ‘save’ for the du part. Let them play around and try to build some intuition before giving hints or solutions.

a. $\int \tan^3 x \sec^5 x dx$

With $u = \sec x$, they should get $\frac{1}{7} \sec^7 x - \frac{1}{5} \sec^5 x + C$.

b. $\int \tan^4 x \sec^4 x dx$

With $u = \tan x$, they should get $\frac{1}{7} \tan^7 x + \frac{1}{5} \tan^5 x + C$.

Look back over the integrals of the form $\int \tan^m x \sec^n x dx$. The approach you take depends on whether m and n are odd or even. Fill in the missing words in the rules below:

- For $\int \tan^m x \sec^n x dx$, if m is _____ and $n \geq 1$ then use $u = \sec x$.
- For $\int \tan^m x \sec^n x dx$, if n is _____ then use $u = \tan x$.

Have a student or group report their answers, which should be: odd; even. Again, if m is odd and n is even, then either substitution choice will work.

There are no clear rules for other integrals of the form $\int \tan^m x \sec^n x dx$, so you have to just try stuff like substitution, integration by parts, and Key Idea 1.3. Calculate each integral. Be persistent! These really can be challenging. If possible, have them continue working on them at home, so they have more time to try things, and come back to them later. A hint is to consider trig identities and integration by parts. Also, there are multiple ways to approach them, so be ready for a student to do something different. Students frequently forget, and so we remind them, that they can always check their answers by differentiating.

a. $\int \sec^3 x dx$

One solution is

$$\begin{aligned} \int \sec^3 x dx &= \int \sec x (1 + \tan^2 x) dx \\ &= \int \sec x dx + \int \sec x \tan^2 x dx \\ &= (\ln |\sec x + \tan x|) + (\sec x \tan x - \int \sec^3 x dx). \end{aligned}$$

So, $2 \int \sec^3 x dx = \ln |\sec x + \tan x| + \sec x \tan x$, and hence

$$\int \sec^3 x dx = \frac{1}{2} (\ln |\sec x + \tan x| + \sec x \tan x) + C.$$

b. $\int \tan^4 x \, dx$

One solution is

$$\begin{aligned}\int \tan^4 x \, dx &= \int \tan^2 x (\sec^2 x - 1) \, dx \\ &= \int \tan^2 x \sec^2 x \, dx - \int \tan^2 x \, dx \\ &= \frac{1}{3} \tan^3 x - \int (\sec^2 x - 1) \, dx \\ &= \frac{1}{3} \tan^3 x - \tan x + x + C.\end{aligned}$$

1.4 Trigonometric Substitution

We begin with a quick review of the basic trig identities and right triangle trig, with the students filling in some of the pieces. We walk them through a step-by-step solution of an example. In that example, we present two methods for setting up the trig substitution - one using a right triangle, and the other using Pythagorean Identities - and encourage the students to use the one they like best. This is followed by a number of integrals for them to calculate.

The final exercise asks the students to develop a general rule for which substitution to use depending on the form of the sum/difference of squares that appears in their integral. As usual, it helps the students to develop to a more personal, deeper understanding if they develop this big-picture view themselves. Our goal is to help them develop a full understanding and intuition, rather than just giving them a list of rules to memorize.

It might help to start this section by explaining that we will be making substitutions using trig functions to integrate functions involving sums and differences of squares, so we begin with a quick review of some Pythagorean and trig identities. Have them work through the first two exercises to review these identities.

(Useful Identities.)

Pythagorean Identities	Trig Identities	Trig Identities
$1 - \sin^2 \theta = \cos^2 \theta$	$\sin \theta = \frac{a}{c} = \frac{\text{opp}}{\text{hyp}}$	$\csc \theta = \boxed{}$
$\sec^2 \theta - 1 = \tan^2 \theta$	$\sec \theta = \frac{c}{b} = \frac{\text{hyp}}{\text{adj}}$	$\cos \theta = \boxed{}$
$\tan^2 \theta + 1 = \sec^2 \theta$	$\tan \theta = \frac{a}{b} = \frac{\text{opp}}{\text{adj}}$	$\cot \theta = \boxed{}$

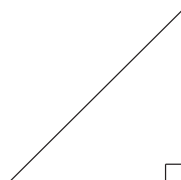


Figure 1.1: Right triangle

On the triangle in Figure 1.1, label the lower left angle θ , the vertical leg a , the horizontal leg b , and the hypotenuse c . In Key Idea 1.4, write down the missing trig identities for $\csc \theta$, $\cos \theta$ and $\cot \theta$.

You should *know* the Pythagorean Identity $\sin^2 \theta + \cos^2 \theta = 1$. What do you get if you divide both sides by $\cos^2 \theta$ and simplify? What if you divide by $\sin^2 \theta$?

The point here is that students don't need to memorize the other two Pythagorean Identities if they know the first and can calculate the other two. Our students seem to not realize this, and appreciate having it pointed out.

You have some options for introducing trig substitutions. You could let the students read and work through Exercise 1.4, then talk about some of the key ideas. You could have them read and work through just the Triangle/Geometry Method first, then give a mini-lecture going over the main points of the Triangle/Geometry Method and contrasting it with a more thorough discussion of the Algebra Method. You could give a mini lecture leading them through both methods in Exercise 1.4. Here are some points to make about each method:

Triangle/Geometry Method: The point of setting up a triangle is to find a relationship between the variable, say x , in the integral, and a new variable, θ . The triangle must satisfy the Pythagorean Theorem, and reflect the sum/difference of squares found in the integrand on either a leg or the hypotenuse. The goal is to set up the triangle in such a way that one of the trig identities is of the form $\sin \theta = x/a$, $\tan \theta = x/a$, or $\sec \theta = x/a$ for some

constant a , because those are all nice substitution equations to work with, and, while other choices will work, with these we won't pick up negative signs from the dx when we go to substitute. Finally, from the triangle, we can use another trig identity to translate the sum/difference found in the integrand into a function of θ . So, there are two key pieces that come out of this process: the substitution equation and the sum/difference relation.

Algebra Method: Again, our goal is to find a relationship between x and θ , but this time we'll do that by looking for a Pythagorean identity (instead of a triangle) that models the sum/difference found in the integrand. If the integrand is a sum, $x^2 + 1$, then we match it with $\tan^2 \theta + 1$ (or, similarly, if the integrand is $x^2 + a^2$, then we multiply the identity by a^2 and match it with $a^2 \tan^2 \theta + a^2$). If the integrand is the difference $1 - x^2$, then we match it with $1 - \sin^2 \theta$, and if the integrand is the difference $x^2 - 1$, then we match it with $\sec^2 \theta - 1$. From this matching, we get both the substitution equation and the sum/difference relation. For example, if the matching is $x^2 + a^2$ with $a^2 \tan^2 \theta + a^2$, then $x^2 = a^2 \tan^2 \theta$ giving a substitution equation of $x = a \tan \theta$ and, from the Pythagorean identity, a sum/difference relation of $x^2 + a^2 = a^2 \sec^2 \theta$.

Note that with either method, it is more efficient (or at least cleaner) to find the sum/difference relation first and use it when substituting, rather than to substitute for x directly and then try to simplify the integrand using algebra and Pythagorean identities. This is why we emphasize finding the sum/difference relation initially. However, some students will still use the later approach.

Finally, we find it helpful to stress that with either method (and as with any substitution), we must differentiate the substitution equation to find a relationship between dx and $d\theta$. After that, we substitute in the integral and carry on just as we would with a u -substitution.

We will now use the identities and a "trig substitution" to integrate $\int \frac{1}{x^2 + 1} dx$. Of course, we know the answer should be $\arctan x + C$, but trig substitution gives us a way to *calculate* it instead of having to just *know* it.

We begin by setting up a substitution equation. There are two methods for doing this; we'll look at both.

Triangle/Geometry Method: Draw a right triangle like the one in Figure 1.1. Since our integral contains a sum of squares $x^2 + 1 = x^2 + 1^2$, label the vertical leg of the triangle x , the horizontal leg 1, and the lower left angle θ . Using the Pythagorean Theorem, $a^2 + b^2 = c^2$, your hypotenuse should be $c = \sqrt{x^2 + 1}$. Notice that $c^2 = x^2 + 1$, matching the sum of squares in our integral.

Next, write down what you get when you apply the Trig Identities in Key Idea 1.4 to your triangle. Which one gives the nicest way to write x as

a function of θ , i.e., in terms of $\sin \theta$, $\sec \theta$, or $\tan \theta$?

You should get that $\tan \theta = \frac{x}{1} = x$. We sometimes tell the students what their answer should have been. This helps them stay on track. We have found that this rarely circumvents them doing the work themselves.

This is our substitution equation:

$$\tan \theta = x.$$

Note also that, from the triangle, we have $\sec \theta = \sqrt{x^2 + 1}$, and hence

$$\sec^2 \theta = x^2 + 1.$$

Thus, using the Triangle/Geometry Method, we have the substitution equation $\tan \theta = x$ and the relation $\sec^2 \theta = x^2 + 1$. Next, we will consider a different method to determine the same equations.

Algebra Method: Notice that $x^2 + 1$ is of the form (variable)² + 1, and, considering the Pythagorean Identities in Key Idea 1.4, $\tan^2 \theta + 1$ is of the same form. So we want $x^2 + 1 = \tan^2 \theta + 1$. Solve the equation for x . Did you get the same substitution equation, $x = \tan \theta$, as with the Triangle/Geometry Method?

Also, knowing that $x^2 + 1 = \tan^2 \theta + 1$, notice that the identity $\tan^2 \theta + 1 = \sec^2 \theta$ can be rewritten as

$$x^2 + 1 = \sec^2 \theta.$$

Thus, using the Algebra Method, we get the same the substitution equation and relation as with the Triangle/Geometry Method.

From either Method, we end up with $x = \tan \theta$ and $x^2 + 1 = \sec^2 \theta$. Before we can perform the substitution, just like with u -substitutions, we need to calculate dx . Using $x = \tan \theta$, calculate $dx = \underline{\hspace{2cm}}$. Substituting, simplifying, and integrating, we get

$$\int \frac{1}{x^2 + 1} dx = \int \frac{1}{\sec^2 \theta} \cdot \sec^2 \theta d\theta = \int 1 d\theta = \theta + C.$$

Finally, as with any substitution method, we need to undo the substitution. Using the transition rule on $x = \tan \theta$, we get $\arctan x = \theta$. Thus,

$$\int \frac{1}{x^2 + 1} dx = \arctan x + C.$$

This is exciting! You no longer need to *memorize* a formula for $\int \frac{1}{x^2 + 1} dx$ because you can *calculate* it instead!

It helps to do an example together on the board before having them work on Exercise 1.4. We suggest $\int \frac{dx}{\sqrt{25-x^2}}$. As you work through the example, invite the students to guide/suggest each step, rather than telling them what to do. This will allow you to help them with their missteps. It will help them figure out the steps and trouble-shoot when they are on their own. It can also be helpful to talk through the choices being made and the difference between an awkward *choice* and a *mistake*. For instance, with the triangle strategy, putting the 5 on a leg instead of the hypotenuse is a *mistake* because it will not result in $\sqrt{25-x^2}$ on any side of the triangle, whereas putting the x on the adjacent leg instead of the opposite leg is simply an *awkward choice* because of the negative sign in the resulting dx . The substitution is $x = 5 \sin \theta$ and the answer is $5 \arcsin(x/5) + C$.

Use similar trig substitutions and whichever method you prefer to calculate each integral. It is always possible to use one of the substitutions $x = \sin \theta$, $x = \sec \theta$, or $x = \tan \theta$, or similar substitutions with constant coefficients (though others may work as well). You might have to play around some to figure out a correct substitution, triangle, or identity.

Depending on the level and independence of our students, we sometimes give them a little time to think about part a. before giving a mini lecture on a solution to the class. They often struggle initially with trig substitution, so some extra help is generally appreciated.

a. $\int \frac{dx}{(x^2 + 5)^{3/2}}$

Answer: $\frac{x}{5\sqrt{x^2+5}} + C$.

Note: The substitution is $x = \sqrt{5} \tan \theta$. The students will need to deal with the cube of the square root, and the fact that 5 is not a perfect square. Although it is a square in the sense that $5 = (\sqrt{5})^2$.

b. $\int \frac{dx}{x^2 \sqrt{16-x^2}}$

Answer: $\frac{-\sqrt{16-x^2}}{4x} + C$.

Note: The substitution is $x = 4 \sin \theta$. The students will need to figure out what to do with the extra x^2 in the denominator.

c. $\int \frac{1}{x\sqrt{x^2-4}} dx$

Answer: $\frac{1}{2} \operatorname{arcsec} \frac{x}{2} + C$

Note: The substitution is $x = 2 \sec \theta$. The students will need to deal with the extra x in the denominator. After substituting and simplifying, they will have $\int \frac{1}{2} d\theta$, which they may have trouble integrating. Then, they won't be sure how to undo the substitution for their answer

$\frac{1}{2}\theta + C$. You might have to point out that they can solve the substitution equation $x = 2 \sec \theta$ for θ .

d. $\int \frac{dx}{\sqrt{4x^2 - 25}}$

Answer: $\frac{1}{2} \ln \left| \frac{2x}{5} + \frac{\sqrt{4x^2 - 25}}{5} \right| + C$

Note: The substitution is $x = \frac{5}{2} \sec \theta$. The students will have to figure out how to get the $4x^2$ as the first term in the radicand. After substituting and simplifying, they'll be integrating $\int \frac{1}{2} \sec \theta d\theta$ which requires a result in Key Idea 1.3 in Section 1.3.

e. $\int \frac{x^2}{\sqrt{9 - x^2}} dx$

Answer: $\frac{9}{2} \arcsin \frac{x}{3} - \frac{1}{2} x \sqrt{9 - x^2} + C$

Note: The substitution is $x = 3 \sin \theta$. After substitution, they are integrating $\int 9 \sin^2 \theta d\theta$, which can be done using integration by parts or using a half-angle identity (see Section 1.3). If they use a half-angle identity, they will end up with an answer involving the angle 2θ instead of θ . In Section 1.3, Exercise 1.3 had them use a Double-Angle Identity to rewrite in terms of the angle θ . This will allow them to un-do the trig substitution using their triangle or Pythagorean Identities. If you don't want to go into this, suggest integration by parts once they get to $\int 9 \sin^2 \theta d\theta$, remove the integral from the assigned list, or assign it as a challenge.

Let's review what we've just done. What is the pattern for trig substitutions? For each expression in Key Idea 1.4, fill in the trig substitution (e.g. $x = a \tan \theta$) you would use if you see the expression in your integral. Also fill in the appropriate identity or triangle, whichever you prefer to use.

For example, in the first column we would expect $x = a \sin \theta$, and either a triangle with hypotenuse a and opposite leg x or the identity $1 - \sin^2 \theta = \cos^2 \theta$.

(Trig Substitutions.)

Expression	$a^2 - x^2$	$a^2 + x^2$	$x^2 - a^2$
Trig Substitution			
Identity or Triangle			

This is often the most difficult section for students, and may require more help from the instructor, either in the form of mini-lectures or increased hints.

We begin with adding fractions to help them recall the algebraic principles behind decomposition, and to make explicit that decomposition is just addition in reverse. We lead them through three forms of decomposition: linear factors, irreducible quadratic factors, and repeated linear factors. At the end of each, we give the students the general form, or have them fill in a piece of the general form.

It may help to begin with a mini lecture setting the stage. Point out that we would like to integrate rational functions (you might remind them of what a rational function is) such as $\int \frac{4x-1}{x^2-x-2} dx$, but none of the tools we have developed so far will work. However, we can factor the denominator and write the integral as $\int \frac{4x-1}{x^2-x-2} dx = \int \frac{4x-1}{(x+2)(x-1)} dx$, which, on it's face, is not helpful. However, there is some algebra we can do which will be helpful. We'll start with the punch-line, to see that it does allow us to integrate, then look at how to get there algebraically.

Consider the equation

$$\frac{3}{x+2} + \frac{1}{x-1} = \frac{4x-1}{(x+2)(x-1)} \quad (1.3)$$

- In Equation 1.3, verify equality by multiplying both sides by the least common denominator, $(x+2)(x-1)$, and simplifying.
- Integrate the left hand side of Equation 1.3. That is, finish computing the integral below:

$$\begin{aligned} \int \frac{4x-1}{(x+2)(x-1)} dx &= \int \frac{3}{x+2} + \frac{1}{x-1} dx \\ &= \int \frac{3}{x+2} dx + \int \frac{1}{x-1} dx \\ &= \end{aligned}$$

The point here is that we integrated the rational function $\frac{4x-1}{(x+2)(x-1)}$ by first decomposing it into the “smaller” and easier-to-integrate rational functions $\frac{3}{x+2}$ and $\frac{1}{x-1}$. This is called partial fraction decomposition. However, decomposing a rational function can be tricky. We will develop general forms for decompositions of certain types of rational functions in the rest of this section.

Let's see how to obtain the decomposition in Equation 1.3. The idea is to determine two fractions, each with one part of the factored denominator and with constants on the top, whose sum is the original rational function. Thus, we are hoping to find constants A and B such that

$$\frac{4x-1}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1}. \quad (1.4)$$

To determine A and B , similar to what we did when verifying Equation 1.3, we multiply both sides by the factored denominator $(x+2)(x-1)$ from the original rational function (this is the least common denominator) to get

$$4x-1 = A(x-1) + B(x+2). \quad (1.5)$$

To solve for the constants A and B , there are two strategies. The first will always work. The second will give you at least some of the constants - if any are left unknown, go back and use the first strategy.

We suggest giving the students time to work through each strategy, then giving a mini lecture on each. In Strategy 1, stress that the system of equations come from matching coefficients of like terms. Of course, there are many ways of solving the system of equations, so students can use whatever approach they like. In Strategy 2, they can choose any value for x , but it's most efficient to choose strategically to make zeros appear. But even if zeros don't appear, they'll still end up with a system of equations from which they may be able to solve for the letters.

Strategy 1: Simplify the right side of Equation 1.5 and collect like terms:

$$\begin{aligned} 4x-1 &= A(x-1) + B(x+2) \\ &= Ax - A + Bx + 2B \\ &= (A+B)x + (-A+2B). \end{aligned}$$

Set the coefficients of like terms from each side of the resulting equation

$$4x-1 = (A+B)x + (-A+2B)$$

equal to each other:

$$\begin{aligned} \text{coefficients of } x: \quad 4 &= A+B \\ \text{constant terms:} \quad -1 &= -A+2B. \end{aligned}$$

Now, solve this system of equations for A and B .

Strategy 2: Choose convenient values of x in Equation 1.5 to make parts become zero:

$$x = 1: \quad 4 - 1 = A(1 - 1) + B(1 + 2) \text{ becomes } 3 = 3B, \text{ so } B = 1.$$

Now, try this using $x = -2$ to determine the value of A .

Hopefully using both methods you obtained $A = 3$, $B = 1$, and hence the decomposition $\frac{4x-1}{(x+2)(x-1)} = \frac{3}{x+2} + \frac{1}{x-1}$.

More generally, the steps for partial fraction decomposition are:

1. Factor the denominator.
2. Determine the fractions in your decomposition, like in Equation 1.4.
3. Solve for values of the unknowns using Strategies 1 and/or 2.
4. Integrate the resulting, simpler, rational functions.

Calculate each integral using partial fraction decomposition.

a. $\int \frac{-2x+5}{x^2+x-2} dx$

Solution:
$$\int \frac{-2x+5}{(x+2)(x-1)} dx = \int \frac{-3}{x+2} + \frac{1}{x-1} dx$$

$$= -3 \ln|x+2| + \ln|x-1| + C.$$

b. $\int \frac{2}{2x^2+x} dx$

Solution:
$$\int \frac{2}{x(2x+1)} dx = \int \frac{2}{x} - \frac{4}{2x+1} dx$$

$$= 2 \ln|x| - 2 \ln|2x+1| + C$$

Note that $\int \frac{-4}{2x+1} dx$ requires a u -substitution, though students tend to make the mistake of just writing down $-4 \ln|2x+1|$ instead.

In the previous exercises you decomposed into fractions with *linear* denominators, i.e., denominators of the form $ax + b$. Each of the resulting fractions had only a constant in the numerator. In other words, we have the following rule:

For a linear factor of $ax + b$ in the denominator, use $\frac{A}{ax+b}$ in your decomposition.

Note: You may use any letters in place of A , such as B or C , as long as you don't use the same letter twice in your decomposition.

Next, consider the decomposition

$$\frac{3x^2 - x + 3}{x^3 + x} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}. \quad (1.6)$$

- a. Notice that the numerator of the left-hand fraction is a quadratic polynomial, while the denominator is a reducible cubic. Characterize the numerators and denominators of the right-hand fractions as: constant, linear, reducible quadratic, irreducible quadratic, or cubic. *You might need to remind the students of terms such as reducible/irreducible, quadratic, and cubic.*
- b. Factor $x^3 + x$. Do the factors match the denominators of the decomposition fractions in Equation 1.6?
- c. To calculate $\int \frac{3x^2 - x + 3}{x^3 + x} dx$, first decompose $\frac{3x^2 - x + 3}{x^3 + x}$ using Strategies 1 and/or 2 to solve for the A , B , and C in Equation 1.6.
- d. Next, use your decomposition to integrate $\int \frac{3x^2 - x + 3}{x^3 + x} dx$.

The answer is $3 \ln|x| - \arctan x + C$.

In the previous example you decomposed into a fraction with a linear denominator, and another with an *irreducible quadratic* denominator. For the fraction with the irreducible quadratic denominator, you used a *linear* numerator $Bx + C$. This leads to the following rule:

For an irreducible quadratic factor $ax^2 + bx + c$ in the denominator, use

$$\frac{Ax + B}{ax^2 + bx + c}$$

in your decomposition.

Students often ask why $\frac{Ax + B}{ax^2 + bx + c}$ instead of, for example $\frac{A}{ax^2 + bx + c}$. The answer is that since the numerator is unknown, we need the most general form it could take for which the fraction would be reduced. Hence, a linear numerator if the denominator is quadratic. Invite them to investigate what would happen if they try using a constant instead of a linear numerator. In the case of equation 5.4 above, with $\frac{3x^2 - x + 3}{x^3 + x} = \frac{A}{x} + \frac{B}{x^2 + 1}$, we can solve for A and B with no problems. However, in other cases, such as $\frac{4x^2 - 2x + 3}{x^3 + x} = \frac{A}{x} + \frac{B}{x^2 + 1}$, we would get a system of equations with no

solution (it would include both $A = 4$ and $A = 3$).

Note: The letters don't matter, only that you don't use the same letter twice in your decomposition.

As a third example, consider

$$\frac{5x+9}{x^3+6x^2+9x} = \frac{5x+9}{x(x+3)^2} = \frac{A}{x} + \frac{B}{x+3} + \frac{C}{(x+3)^2} \quad (1.7)$$

- a. Factor $x^3 + 6x^2 + 9x$. Does your answer agree with Equation 1.7?
- b. Calculate the sum of the three decomposition fractions on the right in Equation 1.7 by first finding the *least* common denominator.

Some students will forget that they want the least common denominator, and will multiply both sides by $x(x+3)(x+3)^2$ instead. They often make this mistake when faced with a repeated factor, and while it's not wrong to make this choice, it does lead to additional work for them to do.

Did you get a fraction with denominator $x(x+3)^2$? If not then find, fix, and make note of your mistake!

- c. In Equation 1.7, notice that the first two decomposition fractions (with A and B) have linear denominators and constant numerators, matching the pattern in Equation 1.3. However, the third decomposition fraction has a *repeated linear* factor in the denominator, but still has a *constant* numerator. The degree of the numerator is determined by the linear factor $(x+3)$, and not by the fact that it is a repeated factor.

Students will ask why not $\frac{Cx+D}{(x+3)^2}$? The extra variable will result in a system of 3 equations in 4 unknowns, so we will end up with C as a free variable. We only need one solution, so we might as well let $C = 0$. Another thing they'll do is to use $\frac{Cx+D}{(x+3)^2}$ term and drop the $\frac{B}{x+3}$ term. In order to integrate this, we would then need to separate into two terms, so we might as well start with two. We encourage the students to investigate these approaches on their own, and sometimes point out that the typical decomposition choices save them effort and time.

- d. Use the decomposition in Equation 1.7 and Strategy 1 and/or 2 to calculate $\int \frac{5x+9}{x^3+6x^2+9x} dx$.

Strategy 2 will give them some of the unknown letters, but not all. They will have to use a system of equations, eventually, with either strategy.

The answer is $\ln|x| - \ln|x+3| - \frac{2}{x+3} + C$.

Considering Equation 1.7, fill in the missing numerators in the rule:

For a repeated linear factor $(ax + b)^2$ in the denominator, use

$$\frac{\boxed{}}{ax + b} + \frac{\boxed{}}{(ax + b)^2} \text{ in your decomposition.}$$

Can you generalize your rule for a linear factor repeated n times? Fill in the missing numerators.

For a repeated linear factor $(ax + b)^n$ in the denominator, use

$$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \frac{\boxed{}}{(ax + b)^3} + \cdots + \frac{\boxed{}}{(ax + b)^n} \text{ in your decomposition.}$$

Summarizing the three examples of partial fraction decompositions, we get the following general rules.

(Partial Fraction Decomposition Rules.)

1. For a **linear factor** of $ax + b$ in the denominator, use $\frac{A}{ax + b}$ as a term in your decomposition.
2. For an **irreducible quadratic factor** of $ax^2 + bx + c$ in the denominator, use $\frac{Ax + B}{ax^2 + bx + c}$ as a term in your decomposition.
3. For a **repeated linear factor** of $(ax + b)^n$ in the denominator, use $\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_n}{(ax + b)^n}$ as part of your decomposition.

Note: We could make other rules, but these rules make our lives easiest when we go to integrate! You might think about what other possible decompositions we could use, and how they relate to these rules.

Use partial fraction decomposition to calculate each integral.

a. $\int \frac{2}{x^3 + 4x^2 + 4x} dx$

Answer: $\frac{1}{2} \ln|x| + \frac{-1}{2} \ln|x + 2| + \frac{1}{x + 2} + C.$

Note that they'll need a u -substitution to integrate the last summand.

b. $\int \frac{x^2 + 2x}{x^4 + x^2} dx$

Answer: $2 \ln|x| - \ln(x^2 + 1) + \arctan x + C.$

Note: $\frac{x^2 + 2x}{x^4 + x^2} = \frac{2}{x} + \frac{-2x + 1}{x^2 + 1}$. For the last summand, students may be unsure how to integrate. Notice that the integral is

$$\int \frac{-2x + 1}{x^2 + 1} = \int \frac{-2x}{x^2 + 1} dx + \int \frac{1}{x^2 + 1} dx.$$

c. $\int \frac{x^3 + 2x^2 - 3x - 18}{x^2 - x - 6} dx$

Hint: first reduce the rational function using long division or synthetic division.

Answer: $\frac{1}{2}x^2 + 3x + \frac{18}{5} \ln|x-3| + \frac{12}{5} \ln|x+2| + C$

Note that students will probably need another reminder of how to divide polynomials and how to then rewrite the rational function. You could repeat what you did in the first section - first an integer long division problem, with a discussion of place value and the analogy that 5 x 's is like 5 tens - or skip the integer long division this time and go straight to the polynomial division. For this exercise, they should get

$$\frac{x^3 + 2x^2 - 3x - 18}{x^2 - x - 6} = (x+3) + \frac{18/5}{x-3} + \frac{12/5}{x+2}.$$

Another option, if you want to avoid the long division issue, is to have them integrate the remainder portion only, i.e. $\int \frac{6x}{x^2 - x - 6} dx$ instead. Then the answer is $\frac{18}{5} \ln|x-3| + \frac{12}{5} \ln|x+2| + C$.

1.6 Integration Practice

It's helpful for students to face a 'random' collection of integrals, without hints as to which technique to use. That's all that is in this section. The exercises could be done in class, or assigned as homework depending on time constraints. We usually do not give our students a lot of in-class time to work on the integration practice, so if it is done in class, the module will take more than our estimated eight days of class to complete. We also rarely hand out answers unless a student asks for a specific integral explicitly.

Calculate each integral.

a. $\int e^{2y} \cos 5y dy$

Answer: $\frac{5}{29} e^{2y} \sin 5y + \frac{2}{29} e^{2y} \cos 5y + C$

b. $\int \frac{1}{\sqrt{9-x^2}} dx$

Answer: $\arcsin x/3 + C$

c. $\int \frac{x}{\sqrt{9-z^2}} dz$

Answer: $-\sqrt{9-z^2} + C$

d. $\int \frac{x^2}{\sqrt{9-x^2}} dx$

Answer: $\frac{9}{2} \arcsin x / 3 - \frac{1}{2} x \sqrt{9-x^2} + C$

e. $\int \sin^3 \theta d\theta$

Answer: $\frac{1}{3} \cos^3 \theta - \cos \theta + C$

f. $\int \sin^4 \theta d\theta$

Answer: $\frac{3}{8} \theta - \frac{1}{8} \sin 2\theta + \frac{1}{8} \sin 4\theta + C$

g. $\int \frac{1}{3x-1} dx$

Answer: $\frac{1}{3} \ln |3x-1| + C$

h. $\int \frac{y-1}{y^2+2y-3} dy$

Answer: $\ln |y+3| + C$

i. $\int \frac{-8}{v^2+2v-3} dv$

Answer: $2 \ln |v+1| - 2 \ln |v-3| + C$

j. $\int \frac{11-7x}{(x^2+2x-3)(x-1)} dx$

Answer: $2 \ln |x+3| - 2 \ln |x-1| - \frac{1}{x-1} + C$

k. $\int \frac{4t^3+2t-2}{t^2(t^2+2)} dt$

Answer: $\frac{1}{t} + \ln |t| + \frac{3}{2} \ln(t^2+2) + \frac{\sqrt{2}}{2} \arctan \frac{t}{\sqrt{2}} + C$

l. $\int \frac{1}{x\sqrt{x^2+3}} dx$

Answer: $\frac{1}{\sqrt{3}} \ln \left| \frac{\sqrt{x^2+3}-\sqrt{3}}{x} \right| + C$

m. $\int w \ln w dw$

Answer: $\frac{1}{2} w^2 \ln w - \frac{1}{4} w^2 + C$

n. $\int \frac{\ln x}{x} dx$

Answer: $\frac{1}{2} (\ln x)^2 + C$

o. $\int \frac{\ln x}{x^2} dx$

Answer: $-\frac{1}{x} \ln x - \frac{1}{x} + C$

p. $\int \arcsin \rho \, d\rho$

Answer: $\rho \arcsin \rho + \sqrt{1 - \rho^2} + C$

q. $\int \sec^3 \phi \tan^3 \phi \, d\phi$

Answer: $\frac{1}{5} \sec^5 \phi - \frac{1}{3} \sec^3 \phi + C$

r. $\int \sec^4 x \tan^3 x \, dx$

Answer: $\frac{1}{6} \tan^6 x + \frac{1}{4} \tan^4 x + C$

s. $\int \sin^3 x \cos^3 x \, dx$

Answer: $\frac{1}{4} \sin^4 x - \frac{1}{6} \sin^6 x + C$

t. $\int \frac{1}{9t^2 - 4} \, dt$

Answer: $\frac{1}{6} \ln \left| \frac{3t - 2}{\sqrt{9t^2 - 4}} \right| + C$